PV Junction Box Incoming Inspection Record

PV IQC INCOMING INSPECTION RECORD · PV-IQC-FM-003 · REV A1

Document No.	Revision	Effective	Retention
PV-IQC-FM-003	A1	2026-10-01	3 years

1. Incoming Information

Supplier:		Material:	PV Junction Box
Model:		Lot No.:	
Quantity:		Inspection Date:	
Inspector:		Reference:	PV-IQC-WI-003

2. Sampling

Lot size: _____ Sample size: _____ AQL: Major 0.65 / Minor 1.5 Inspection level: General II

3. Inspection Items and Results

Item	Requirement	Measured	Result
Appearance	No deformation, crack, burr or obvious color difference		Pass / Fail
Dimensions	Length, width, height and mounting holes per drawing ±0.5 mm		Pass / Fail
Diode forward voltage	≤0.45 V at 1 A	_____ V	Pass / Fail
Diode reverse leakage	≤10 μA at rated reverse voltage	_____ μA	Pass / Fail
Terminal pull force	≥40 N, no pull-out or looseness	_____ N	Pass / Fail
Insulation resistance	≥100 MΩ at 500 VDC	_____ MΩ	Pass / Fail
Dielectric withstand	2.5 kV AC / 1 min, no breakdown or flashover		Pass / Fail
Ingress protection	IP68 per technical agreement		Pass / Fail
Flame retardancy	750°C glow wire, no sustained flame		Pass / Fail
Marking / packaging	Model, polarity and lot clear; moisture and impact protection		Pass / Fail

4. Defect Record

Location: _____ Description: _____ Quantity: _____ Class: ■ CR ■ MAJ ■ MIN

5. Lot Disposition

■ Accept ■ Accept by deviation ■ Reject and return ■ Sort and re-inspect
Disposition note: _____

Quality Engineer: _____ Date: _____

Supplier Representative: _____ Date: _____

Note: file this record with the diode datasheet, IP test report, flame-retardancy report and defect photos. Isolate the lot immediately if a critical defect is found.